Trials

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Recruitment strategies for a randomized controlled trial in a rural American Indian community: the Cooking for Health Study

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Abstract

Background Conducting diet-focused interventions in rural, American Indian (AI), and/or under-resourced communities is critical to improve health. However, interventions that may work in urban or suburban settings may not work in rural or under-resourced communities due to differences in types of locally available food, food preferences, food insecurity, and/or limited access to grocery stores or supermarkets. This paper describes the acceptability of recruitment strategies used in the Cooking for Health (CFH) study, a culturally targeted healthy food budgeting, purchasing, and cooking skills intervention for AIs with type 2 diabetes who live on or around a large reservation in the North-Central United States.

Methods CFH was a two-armed parallel group randomized controlled trial (RCT) designed to evaluate the effectiveness of a 12-month intervention on diet, healthy food self-efficacy, food budgeting and cooking skills, and healthy food purchases. The intervention consisted of monthly lessons and videos related to cooking healthy foods, food budgeting skills, and optimal nutrition for diabetes management. AI men and women 18 + years old with diagnosed type 2 diabetes were eligible to participate. Recruitment strategies included passive, active, and word-of-mouth (refer-a-friend) methodologies. Additionally, research staff worked closely with a community partner, the Tribal Diabetes Program, to facilitate enrollment.

Results Of the 276 individuals screened for eligibility for the study, 226 individuals were deemed eligible to participate. Of these, 176 enrolled, 25 individuals scheduled a baseline study visit but did not show up, and 25 individuals declined participation. Reasons for declining participation included: no longer interested; too busy/employed full-time; already managing diabetes with own healthcare provider; or moved out of the area.

Conclusions These results support a multi-strategic recruitment approach of passive, active and word-of-mouth strategies for successful recruitment and enrollment of AIs in diet-focused RCTs. It is likely that the recruitment successes were largely due to community involvement and ownership in the study. The importance of trusting mutually beneficial and shared decision-making is also critical to project success.

Trial registration ClinicalTrials.gov, TRN: NCT03699709. Registered on 5 October 2018, https://clinicaltrials.gov/study/NCT03699709.

Keywords American Indian, North American, Diabetes mellitus, Rural population, Randomized controlled trial, Education, Nutrition, Cooking, Budgeting

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Background

Conducting diet interventions in rural, American Indian (AI), and/or under-resourced communities is critical to improve health outcomes. However, diet interventions that may work in urban or suburban settings may not work in rural or under-resourced settings due to differences in the type of food available locally, food preferences, food insecurity, and/or limited access to grocery stores or supermarkets [1, 2].

Randomized controlled trials (RCTs) are often considered the gold standard for evaluating effectiveness of many types of interventions, including diet interventions [3]. Over the past two decades, numerous obesity, diabetes, and cardiovascular health interventions for adults have been implemented in clinical settings in AI communities, but only a handful of these studies have employed an RCT-design [4–10]. Importantly, many leading research organizations (e.g., National Institutes of Health, United States Food and Drug Association) highlight the importance of racial/ethnic diversity in RCTs. However, AIs continue to be underrepresented in clinical research when compared to other racial/ethnic groups in the US. Among successful RCTs that involve AI communities, strategies for successful recruitment of AIs are largely unreported. In one literature review focused on common recruitment strategies in health-related trials in AI communities, passive (advertisements), individual-level (direct mailings, phone calls) and relational (healthcare/friend referrals) strategies were commonly utilized [11]. Unfortunately, due to the limited number of trials in AI communities, the review summarized strategies by combining approaches across all available RCTs ($n=41$) that involve AIs in the US—with topics as discordant as trauma care or sexual health to cancer screening or management of cardiovascular diseases. Given vast differences in study populations by geography, culture, age, and underlying health status, the utility of the findings is unclear. Additionally, half of the RCTs were conducted more than 10 years ago, and Americans have changed how they interact with healthcare, research, and technology during this time—particularly post COVID-19 pandemic.

The Cooking for Health (CFH) Study was an academic-tribal partnership to develop, implement, and test the effectiveness of a culturally targeted healthy food budgeting, purchasing, and cooking skills intervention for AIs with type 2 diabetes who live on or around a large reservation in the North-Central United States (US). The premise of the project was based on a series of focus groups and key informant interviews that highlighted several barriers to healthy eating for adults with type 2 diabetes including (1) difficulty budgeting for food on low-incomes; (2) low health literacy when purchasing food (e.g., inability to use in-store scales to convert foods priced "per pound" to total dollar costs); and (3) limited cooking skills [12–14].

Details of CFH have been described previously [15]. In brief, the intervention focused on AI adults with type 2 diabetes who report doing most of their household's food shopping and/or cooking. In total, there were 176 participants randomized to one of two arms. The intervention arm participated in a 12-month food budgeting, purchasing, and cooking skills intervention. The control arm received 12 monthly newsletters during the intervention period and access to all intervention materials at the end of the study. For all participants, three in-person data collection visits occurred at baseline, month 6 and month 12. Each visit included collection of physical measurements, such as height, weight and blood pressure, biomarkers, and completion of surveys on psycho-social, behavioral, and diet factors. A subset of participants randomized to the intervention arm were invited to take part in semi-structured phone interviews after the month 6 and month 12 visits to assess satisfaction with the intervention and to provide feedback on the study curriculum.

The purpose of this paper is to describe the acceptability of recruitment strategies for CFH.

Methods

Inclusion/exclusion

Recruitment began in February 2020. At that time, eligibility criteria included the following: self-identify as AI; 18–60 years old; diagnosed type 2 diabetes (confirmed through medical record review of physician diagnosis and/or use of oral diabetic medications or insulin or with a fasting plasma glucose equal or greater to 126 mg/dl or HbA1c equal or greater to 6.5% as documented in medical records); reside on the reservation where the study was being conducted; and self-identify as a person who holds most of the household responsibility for food budgeting, shopping, and/or cooking. Participants who were pregnant or with underlying morbidity that may impact diet (i.e., history of bariatric surgery; chronic kidney disease; dialysis) or ability to fully participate in the intervention (i.e., houseless; cognitively impaired; another household member participating in the intervention) were excluded. This inclusion/exclusion criteria maximized both internal validity and statistical power and minimized risk of contamination. For example, given that the intervention focused on increasing the number of healthy foods purchased and/or cooked, individuals without a reliable place to cook or store food were excluded from participation since access to a functional stove, cooktop, or microwave was needed to complete the intervention. Likewise, only one person per household was eligible to participate

to avoid contamination due to discordance of study arm (i.e., intervention arm or control arm) within a household. For the age window, we chose 18–60 years old as the target age range since mid-life is when complications due to diabetes often arise (e.g., chronic kidney disease, dialysis, cardiovascular disease). Eligibility screening was done by phone, or in-person including on-site at local health fairs and community gatherings.

During the study period, small changes were made to the eligibility criteria to broaden the reach of the study and to increase enrollment. In mid-2021, the age requirement was relaxed to accommodate individuals greater than 60 years old with no upper age limit. We chose to relax the upper age limit because during the recruitment period, we recognized that there were community members with diabetes outside of the age-range who met all other eligibility criteria and were eager to participate. Before relaxing the age window, we consulted with clinical staff (i.e., physician, dietitian) in the community—who felt that age would not influence ability to engage with the intervention or the primary outcomes of interest for the study (i.e., change over time in dietary metrics, healthy food self-efficacy, food budgeting and cooking skills, and healthy food purchases) [15]. As such, to maximize reach, we decided to relax the upper age criteria. Then, in mid-2022, recruitment slowed down. In conversations with our community partners, we determined that we may have exhausted the potentially eligible and interested population who live on the reservation ($n=138$). As such, we expanded the catchment area to include AI who resided in towns that border the reservation. The study was closed to recruitment in October 2022.

Recruitment strategies to maximize reach

Recruitment strategies included passive, active and word-of-mouth methodologies. The passive strategies used were newspaper ads, radio announcements, flyers and study contact tear sheets, and social media (i.e., posting the recruitment flyer on a community Facebook page). Additionally, research staff worked closely with our community partner, the Tribal Diabetes Program to facilitate enrollment via referrals. The Tribal Diabetes Program is a Tribally run community-based program on the reservation that provides both educational support and clinical care to adult community members with diabetes to promote optimal disease management. The Tribal Diabetes Program mailed study invitations to patients receiving care at their clinic on the reservation.

Study staff also utilized a variety of active recruitment strategies, including tabling at local community events. Venues for tabling included community health fairs and other highly visible community locations (e.g., lobbies of the Indian Health Service (IHS) Hospital, the local large grocery store, or school events). Flyers that described the study were included in food boxes distributed to AIs experiencing food insecurity and/or AIs with diabetes as part of the United States Department of Agriculture Food Distribution Program on Indian Reservations, the Tribal Youth Diabetes Prevention Program, and the local food pantry. The Tribal Diabetes Program staff, as well as IHS healthcare providers, provided study flyers to patients when they came in for regularly scheduled in-person diabetes care wellness visits.

In September 2021, we instituted a "Refer a Friend" word-of-mouth recruitment method to incentivize current participants to reach out to friends in the community who may be interested in participating in the study. The "Refer-a-Friend" incentive was a $50 payment to show appreciation for the participant's time spent helping to promote the study. Enrolled participants received three numbered referral cards to give to friends. If a participant's friend contacted study staff, screened eligible for the study, enrolled in the study, and turned in the referral card, the participant who gave them the referral card would receive a $50 incentive. The identity of the newly enrolled participant was not disclosed to the participant making the referral. The "Refer a Friend" program was a unique strategy used to help mitigate the impact of the COVID-19 pandemic on in-person recruitment activities.

Results

Recruitment

The targeted enrollment for CFH was 165 participants. There were 276 individuals screened for eligibility by study staff throughout the enrollment period. Of the 276 individuals screened, 226 (82%) individuals were eligible and 50 (18%) individuals were ineligible. The reasons for ineligibility were: (1) did not have type 2 diabetes or unable to determine a diabetes diagnosis ($n=32$); (2) history of bariatric surgery, were on dialysis, or were pregnant ($n=9$); or (3) residing outside of the study catchment area, residing at the same residence as another person who was already enrolled in study, did not identify as American Indian, or became incarcerated after being screened ($n=9$). Of the 226 individuals eligible to participate in the study, 176 enrolled (78%), 25 (11%) scheduled a baseline study visit but did not show up (with no further communication from them), and 25 (11%) individuals declined participation. Of the 25 individuals who declined, reasons for refusal included: (1) no longer interested; (2) too busy or employed full-time; (3) already managing diabetes with their healthcare provider; or (4) moved out of the area. The recruitment, eligibility and enrollment numbers are shown in Fig. 1.

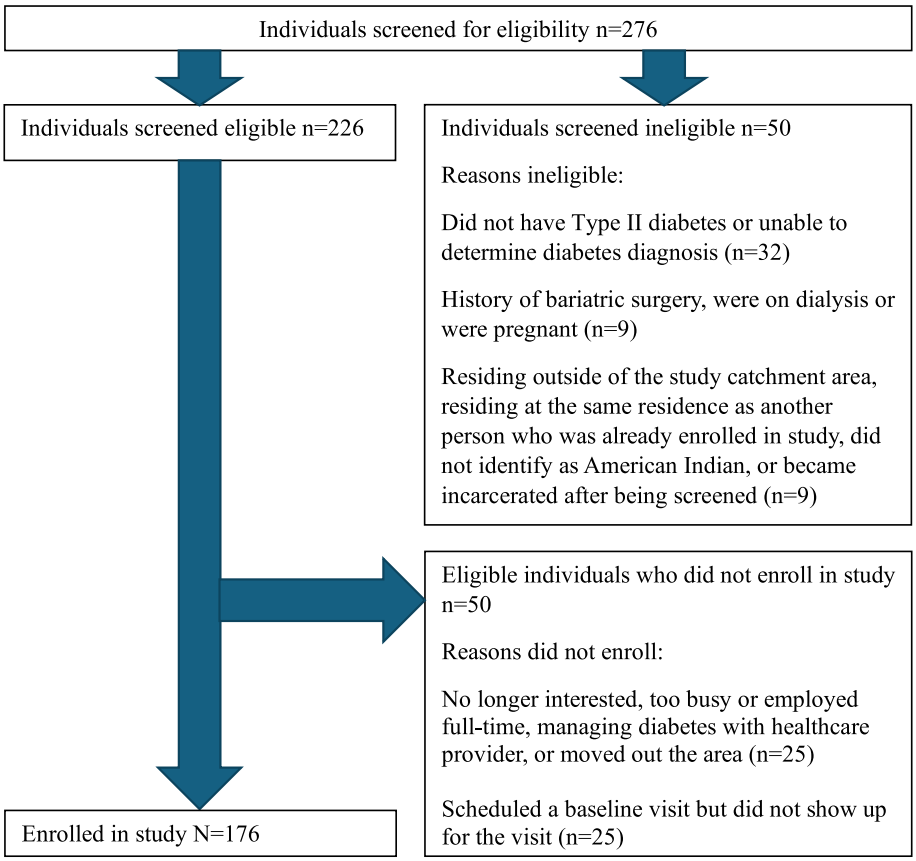


Fig. 1 Recruitment, eligibility, and enrollment numbers

The Tribal Diabetes Program mailed study invitation letters to 500 patients as part of recruitment. The direct mailing yielded a response from 20 individuals of which 16 enrolled in the study. Of the four who did not enroll, three were scheduled for the baseline visit, but did not show up and could not be further reached, and one responded that they were not interested in the study. The "Refer a Friend" incentive payment program yielded 37 enrollees in the study (21% of total enrolled).

As part of screening, individuals were asked to respond to the question: "How did you hear about the study?" Response rates by active and passive recruitment strategies are shown in Table 1.

Impact of COVID-19 on recruitment

We began recruitment in late February 2020. Recruitment was originally projected to take 1 year. From February 27, 2020 through March 10, 2020, ten participants enrolled in the study. Recruitment abruptly came to a halt after the first documented case of COVID-19 in the area on March 10, 2020. Enrollment was paused from March 12, 2020 to April 3, 2021 due to COVID-19 related research restrictions mandated by the University of Washington (UW) Institutional Review Board (IRB) and Tribal Council initiatives to prevent the spread of COVID-19. During the recruitment pause, the reservation followed an emergency response that moved from Level 1 (minimal risk to public health) to Level 5 (critical risk and severe threat to public health) based on 7-day test positivity, 7-day active case average, and hospital capacity.

On April 4, 2021, the Tribal Council downgraded the risk to Level 3 (i.e., moderate risk to public health), and recruitment restarted. To address safety concerns, we implemented a comprehensive COVID-19 pre-visit screening protocol. We contacted participants within 24 h of their scheduled in-person research visits to screen for COVID-19 symptoms and/or exposure. The screening questions were also asked upon the participant's arrival to the clinic. If any COVID-19 symptoms or exposure was reported, the research visit was rescheduled. Delivery of the intervention videos was also modified due to COVID-19. Originally, participants were asked to view the CFH videos at the Tribal Diabetes Clinic or a satellite office. However, due to COVID-19, participants were distributed tablets with hard-downloaded copies of all

Table 1 Response rates by active and passive recruitment strategies ($n=276$)

"How did you hear about the study"	Number of respondents screened
Passive recruitment strategies	
Tribal Diabetes Program invitation letter	20
Poster/flyer	40
Facebook/social media	17
Active recruitment strategies	
Community event	26
Word of mouth	61
Refer-a-friend card	53
Other	16
Unknown (i.e., left question blank)	43

videos. This allowed participants to view all study materials at home without access to the internet—and minimized time at the clinic.

Once recruitment resumed, 50% of participants were enrolled by November 2021, 75% enrolled by April 2022, and the final participants were enrolled in October 2022. In total, 176 participants enrolled in the study (49 males, 127 females; 88 were randomized to the intervention arm and 88 were randomized to the control arm). Both the mean and median age for enrolled participants were 50 years old. There were 86% ($n=152$) participants 18–60 years old and 14% ($n=24$) were 61 + years. Baseline characteristics of enrolled participants are shown in Table 2. Participants in the intervention arm were slightly older (51.0 years in intervention arm and 48.2 years in control arm) and more likely to identify as female (78.4% versus 65.9%). Household size differed slightly across study arms with the intervention arm participants reporting more adults (3.74 versus 2.85), but fewer children (1.56 versus 1.95) in their household than participants in the control arm. There were no differences in education levels across study arm, and household food insecurity was high in both arms (40.9 and 45.5% of study participants in the intervention and control arms, respectively).

The total enrolled sample of 176 participants exceeded the proposed sample size of 165 participants. We chose to enroll an additional eleven participants because there was widespread community interest in the study and the uncertainty of retention due to the ongoing COVID-19 pandemic. This study demonstrated that it was feasible

Table 2 Baseline characteristics of enrolled study participants ($N=176$)

Baseline characteristics of enrolled study participants ($N=176$)	*Control ($N=88$)*		*Intervention ($N=88$)*	
Age, years	48.3	11.2	51.0	11.9
Sex, % female	65.9		78.4	
Education, %				
< High school degree	14.8		14.8	
High school degree	39.8		31.8	
Some college	33.0		37.5	
College grad +	12.5		14.8	
Currently employed, %	43.0		39.2	
Annual income, %				
< $9,600	37.5		37.5	
$9,600-$17,999	19.3		17.0	
$18,000-$39,999	22.7		12.5	
$40,000 +	9.1		13.6	
No. Adults in household	2.9	2.5	3.7	6.8
No. Children in household	2.0	2.1	1.6	1.8
Use of 1 + Federal or Tribal Assistance Programs, %*	96.6		96.6	
BMI, kg/m2	34.6	8.6	34.1	8.0
Food security, %				
High food security	28.4		29.5	
Marginal food security	30.7		25.0	
Food Insecure	40.9		45.5	
HbA1c, %	8.7	2.1	8.4	2.3

* Data are mean (SD), else %

to successfully recruit and enroll AI community members despite multiple barriers, including the COVID-19 pandemic.

Discussion

Successful recruitment strategies

Trusting academic-community partnerships are key to conducting successful research in AI communities. The partnerships should be mutually beneficial with shared power/decision-making and project ownership. In CFH, all components of the study were designed and implemented through the collaboration and partnership of the UW, the Tribal Diabetes Program, and a long-standing research-focused organization on the reservation. All study staff that interacted with participants were community members with deep ties to the community. Previous work supports the importance of involving community members in all components of research, from design to dissemination to maximize success [11, 16]. It is likely that our recruitment successes are largely due to community members' involvement and ownership in the study.

The ability to embed recruitment strategies within ongoing community programs and events helped streamline the recruitment process. For instance, staff at the Tribal Diabetes Program were able to actively recruit patients at the time of their scheduled wellness visits. Study staff presence at local health fairs and community gatherings effectively minimized the need for phone service because interested individuals could be also screened in-person at these events. The ability to provide transportation to and from study visits, sometimes 2 h round trip, enabled participants with limited transportation options to participate. The "Refer a Friend" incentive (word-of-mouth) from those already enrolled in the study was also a successful strategy as it increased the reach to interested individuals and yielded 21% of the number of individuals who enrolled in the study.

Barriers to recruitment

All recruitment materials (e.g., social media and radio advertisement, direct mailings, flyers) included a phone number for interested individuals to call to learn more about the study. However, internet and cellular service were unreliable in some areas of the study catchment area. Additionally, many individuals in the community use disposable phones, pay-as-you-go cell phone plans, shared phone lines, or message-only phones. This rendered frequent communication with interested individuals by phone challenging if phone numbers changed, monthly minutes ran out, or call-back messages were not replied. Also, the community is a rural area that covers 4200 square miles of land; public transportation options are very limited, and some interested individuals needed assistance to get to the study site to attend the study visits.

There were 25 eligible individuals who scheduled baseline visits but did not show up. Because there was no further contact with these individuals, it is not known why they did not show for their scheduled visits. However, it could be speculated that concern of COVID-19, or lack of access to reliable phones or available transportation at the time of the scheduled visit might have been contributing factors.

Due to typical distribution of household labor in the participating community, we a priori expected more females than males to participate in the study since it is more common for women than men to do the household shopping, cooking and budgeting. Of the 276 individuals screened for eligibility. 71% identified as female ($n=196$) and 29% identified as male. Likewise, a similar distribution of females and males enrolled in the study: 72% females versus 28% males.

A major unanticipated barrier to recruitment was a 13-month pause on recruitment due to the COVID-19 pandemic. Some research projects, especially those located in more urban areas, had the ability to pivot to online platforms for eligibility screening, informed consent and intervention delivery using web-based apps such as REDCap [17, 18]. However, it was not feasible for CFH to transition to an online platform because internet services are not widely available.

Research and practice implications

CFH was designed to test the effectiveness of a community-informed cooking and healthy food budgeting intervention on diet quality, healthy food self-efficacy, food budgeting and cooking skills, and healthy food purchases. If the intervention is shown to be effective, the intervention will be incorporated into the existing Tribal Diabetes Program to optimize health among adults with diabetes in the community.

Conclusion

The results reported herein support a multi-strategic recruitment approach of passive, active and word-of-mouth strategies for successful recruitment and enrollment of AIs who live in a rural area into a diet-focused RCT. Careful consideration of eligibility criteria at the design phase and utilizing the least restrictive criteria needed to achieve the aims of the project will ensure optimal reach. Leveraging research staff local to the Tribal community also helps to build community trust and infrastructure to maximize intervention success and sustainability. Flexibility of the study team, partners, and participants during the COVID-19 pandemic was essential to recruitment success. The importance of trusting mutually beneficial and shared decision-making/power among all partners is also critical to project success.

Abbreviations

AI	American Indian
CFH	Cooking for Health
IHS	Indian Health Service
IRB	Institutional Review Board
RCT	Randomized controlled trial
US	United States
UW	University of Washington

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Authors' contributions
AMF, CNH, ANT, LGB, MO, MCB, JU, AH and SAAB made substantial contributions to the conception and design of the work. AMF, RO, CNH, MO, AH, RWB and SAG conducted the research. SAG and AMF wrote the paper and SAG prepared Fig. 1. SAG, AMF, RO, CNH, LGB, ANT and MCB edited the paper. SAG and AMF have primary responsibility for the final content. All authors read and approved the manuscript.

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Data availability
Data described in the manuscript and/or analytic code are not publicly available due to data sharing agreements with the authors and the participating Tribe (the authors are not authorized to share data). Use of the data requires permission from the Tribe and Indian Health Services IRB. Please contact the study team for more information.

Declarations

Ethics approval and consent to participate
Research was performed in accordance with the Declaration of Helsinki. The CFH RCT was approved through the University of Washington IRB (STUDY00004114), Great Plains IRB (GP IRB #18-R-06GP), and the Tribal Health Board. ClinicalTrials.gov, TRN: NCT03699709. Registration date: 5 October 2018, https://clinicaltrials.gov/study/NCT03699709. Participants provided verbal consent during recruitment. Written, informed consent was obtained from participants enrolled in the RCT.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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